

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1) Closest pair - Negative and positive co-ordinates using brute force

Problem Interpretation

Given the set of five points

$$P = \{(-2, -1), (3, 4), (0, 0), (5, 1), (-1, -2)\}$$

The closest pair problem is to find the two points having the minimum Euclidean distance between them

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Given points

Let

$$P_1 = (-2, -1)$$

$$P_2 = (3, 4)$$

$$P_3 = (0, 0)$$

$$P_4 = (5, 1)$$

$$P_5 = (-1, -2)$$

$n = 5$, the number of unique pairs is

$$\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$$

10 pairs will be compared

Distance calculation for all pairs

Q 1: P_1 and P_2

$$d(P_1, P_2) = \sqrt{(3 - (-2))^2 + (4 - (-1))^2}$$

$$= \sqrt{5^2 + 5^2} = \sqrt{50}$$

$$= 7.071$$

Pair 2: P_1 and P_3

$$\begin{aligned}d(P_1, P_3) &= \sqrt{0 - (-2)^2 + 0 - (-1)^2} \\&= \sqrt{2^2 + 1^2} \\&= 2.236\end{aligned}$$

Pair 3: P_1 and P_4

$$\begin{aligned}d(P_1, P_4) &= \sqrt{5 - (-2)^2 + (1 - (-1))^2} \\&= \sqrt{7^2 + 2^2} \\&= \sqrt{53} \\&= 7.280\end{aligned}$$

Pair 4: P_1 and P_5

$$\begin{aligned}d(P_1, P_5) &= \sqrt{(-1 - (-2))^2 + (-2 - (-1))^2} \\&= \sqrt{1^2 + (-1)^2} \\&= \sqrt{2} = 1.414\end{aligned}$$

Pair 5: P_2 and P_3

$$\begin{aligned}d(P_2, P_3) &= \sqrt{(0 - 3)^2 + (0 - 4)^2} \\&= \sqrt{(-3)^2 + (-4)^2} = \sqrt{25} \\&= 5\end{aligned}$$

Pair 6: P_2 and P_4

$$\begin{aligned}d(P_2, P_4) &= \sqrt{(5 - 3)^2 + (1 - 4)^2} \\&= \sqrt{2^2 + (-3)^2} = \sqrt{13} \\&= 3.606\end{aligned}$$

Pair 7: P_2 and P_5

$$\begin{aligned}d(P_2, P_5) &= \sqrt{(-1 - 3)^2 + (-2 - 4)^2} \\&= \sqrt{(-4)^2 + (-6)^2} \\&= \sqrt{52} \\&= 7.211\end{aligned}$$

Pair 8: P_3 and P_4

$$\begin{aligned} d(P_3, P_4) &= \sqrt{(5-0)^2 + (1-0)^2} \\ &= \sqrt{5^2 + 1^2} = \sqrt{26} \\ &= 5.099 \end{aligned}$$

Pair 9: P_3 and P_5

$$\begin{aligned} d(P_3, P_5) &= \sqrt{(-1-0)^2 + (-2-0)^2} \\ &= \sqrt{(-1)^2 + (-2)^2} = \sqrt{5} \\ &= 2.236 \end{aligned}$$

Pair 10: P_4 and P_5

$$\begin{aligned} d(P_4, P_5) &= \sqrt{(-1-5)^2 + (-2-1)^2} \\ &= \sqrt{(-6)^2 + (-3)^2} = \sqrt{45} \\ &= 6.708 \end{aligned}$$

Closest pair

The minimum distance among all the calculated distance is

$$\sqrt{2} \approx 1.414$$

Closest pair is

$(-2, -1)$ and $(-1, -2)$

1.414 units

Number of pairwise comparisons

For n points, the brute force algorithm compares every

unique pair

$$\frac{n(n-1)}{2}$$

For $n=5$

$$\frac{5(4)}{2} = 10$$

Here, pairwise comparisons

10

the complexity analysis

The number of comparisons is

$$\frac{n(n-1)}{2}$$

The grows approximately

$$O(n^2)$$

The time is $O(n^2)$
complexity

The space complexity is

$$O(1)$$

$$O(\log n)$$

Proof of correctness

$$\sqrt{2} = 1.414$$

$$\sqrt{5} = 2.236$$

$$\sqrt{13} = 3.606$$

$$2 < \sqrt{5} < \sqrt{13} < \dots$$

$$\text{best path} = \{(-2, -1), (-1, -2)\}$$

$$\text{Minimum distance} = \sqrt{2}$$

$$\approx 1.414$$